

Multiphysics Modelling of the ALIX Industry 5.0 Production Line

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Abstract

This report presents the current state of the multiphysics digital twin developed for the ALIX Industry 5.0 production line. The modelling effort follows the Modelica/FMU approach to couple mechanical, electrical and thermal domains of three independent subsystems: a TECHTOP TIA56-2 induction-motor-driven conveyor, an XY10 pick-and-place station, and a Dobot CR5 collaborative robot. The conveyor model has been validated against TDMS field measurements at the line and reproduces the measured 50 Hz fundamental phase current within 14 %, the supply voltage exactly, and the steady-state input power (108 W). A thermal extension has been added (lumped-capacity motor body coupled to ambient via a thermal conductance) and verified end-to-end against the Dymola reference to within 0.0001 K. A complete 6-degree-of-freedom multibody model of the CR5 cobot has been derived from CAD-extracted mass-inertia parameters and exported as an FMI 2.0 Co-Simulation FMU. Spec compliance currently stands at approximately 88 % of the explicit deliverables; remaining items are the spatial representation and coupling matrix, scheduled for the next two days. The work is on track for the 30 May semester review.

1. Context and Scope

The ALIX line is a bench-top Industry 5.0 reconfigurable production line operated at the Euler Laboratory. The internship objective, as defined in the original specification, is to build a multiphysics digital twin covering, for each subsystem:

- **mechanical behaviour** (kinematics, dynamics, friction, gearbox reduction);
- **electrical behaviour** (AC supply, motor electromagnetic model, current and power);
- **thermal behaviour** (motor heating, conduction to ambient, temperature evolution);
- **spatial representation** to locate components and identify coupling effects (vibration, thermal gradient, EMI).

The implementation language is Modelica; subsystems are exported as FMU 2.0 Co-Simulation units and orchestrated from Python using the FMPy library. The line-level orchestration represents an Industry-5.0 cyber-physical pattern in which each station runs as a self-contained FMU connected through symbolic part flow and shared electrical bus.

2. Methodology

Each subsystem is modelled in the 3DEXPERIENCE Dymola Behavior-Modeling environment using the standard Modelica 4.1 libraries. Parameters are obtained from three sources:

- **Nameplate data** (motor 0.09 kW / 230 V Y / 50 Hz / 4-pole; gearbox NMRV 030 i = 20);
- **CAD extraction** via FreeCAD for the CR5 cobot (per-link mass and inertia tensors, validated against the manufacturer brochure to 94.9 %);
- **Field measurement** from TDMS recordings (six channels at 6 kHz: two voltages, two currents, two auxiliary).

Validation follows a two-stage protocol: (i) the model is run inside Dymola against analytical predictions (steady-state RMS, energy balance) and against TDMS recordings band-pass filtered at 50 Hz to extract the line-frequency component; (ii) the exported FMU is re-simulated under Python via FMPy and the outputs are compared back to the Dymola reference. Stage (ii) caught one early issue — the original CurrentQuasiRMSSensor produced a flat steady-state value while the supervisor required the full AC waveform; the model was extended with a direct polyphase-pin current readout (`i_phase_inst`) and the FMU re-exported.

3. Subsystem Models

An overview of the three implemented subsystems is given in Table 1. Detailed descriptions follow.

Subsystem	Status	Physics domains covered	FMU outputs
Conveyor	v3 — complete	Mechanical + electrical + thermal	8 (<code>w_motor</code> , <code>tau_shaft</code> , <code>v_belt</code> , <code>x_belt</code> , <code>i_phase</code> , <code>i_phase_inst</code> , <code>P_elec</code> , <code>T_motor</code>)
CR5 cobot	Complete (6-DOF)	Mechanical (multibody, damped)	16 ($6 \times q$, $6 \times \dot{q}$, <code>P_total</code> , <code>x/y/z TCP</code>)
XY10 station	Complete (3 axes + vacuum)	Mechanical + electrical (DC)	11 (positions, currents, vacuum state)

3.1 Conveyor (Conveyor_updated_v3)

The conveyor model couples a three-phase 244 V RMS / 50 Hz sinusoidal supply to an IM_SquirrelCage induction machine ($R_s = 12 \, \Omega$, $R_r = 8 \, \Omega$, $L_m = 0.45 \, \text{H}$, $L_\sigma = 25 \, \text{mH}$, 4-pole). The mechanical chain — motor $\rightarrow$ NMRV 030 worm gearbox ($i = 20$) $\rightarrow$ drum ($r = 25 \, \text{mm}$) $\rightarrow$ belt mass (1 kg) $\rightarrow$ viscous damper (1 N·s/m) — is connected through ideal-gear and ideal-rolling-wheel components.

The thermal extension uses a 1-state lumped-capacity formulation: the motor heat port is connected to a 500 J/K HeatCapacitor, then to ambient (293.15 K) through a 1.5 W/K ThermalConductor. A TemperatureSensor exposes `T_motor` as a new FMU output. The resulting thermal time constant is $\tau = C/G \approx 333 \, \text{s}$ and the asymptotic temperature rise $\Delta T_\infty = P_{\text{loss}} / G \approx 72 \, \text{K}$, consistent with the 108 W electrical input and near-zero mechanical output at light load.

Steady-state numerical results at $t = 60 \, \text{s}$:

- **Motor speed:** 1 499 rpm (slip $\approx 0.04 \, \%$, near synchronous);
- **Shaft torque:** 0.0065 N·m (light-load operation);
- **Belt linear velocity:** 0.196 m/s;
- **Phase current (RMS):** 1.629 A; instantaneous peak $\pm 2.30 \, \text{A}$ (clean 50 Hz sinusoid);
- **Input electrical power:** 108 W;
- **Motor body temperature:** 305.10 K (31.95 °C, rise of 11.95 K from ambient).

FMU-to-Dymola agreement: `T_motor` 0.0001 K, `i_phase` exact to four decimals, `P_elec` 0.0003 W.

3.2 CR5 Collaborative Robot

A 6-DOF Modelica multibody model of the Dobot CR5 cobot has been built from CAD-extracted geometry. Per-link masses, centres of mass and inertia tensors were obtained in FreeCAD with cylinder approximations to guarantee physically valid (positive-eigenvalue) tensors. Total model mass is 23.74 kg, within 5.1 % of the manufacturer's 25 kg specification.

Joint axes follow the standard cobot orthogonal convention (J1, J6 about local Z; J2, J3, J5 about Y; J4 about X). Each revolute joint carries a small viscous damper (J1: 1.0 N·m·s/rad, decreasing to 0.2 N·m·s/rad at the wrist) representing gear and bearing friction. Six Modelica.Mechanics.Rotational.Sources.Torque blocks expose $\tau_{ref}[1..6]$ as FMU inputs.

Verification: starting from joints at zero with zero applied torque, J2 swings under gravity to a peak of $\approx 160^\circ$, settles toward equilibrium near 147° at $t = 3$ s — the expected damped pendulum response. The FMU has 6 inputs, 16 outputs (6 q , 6 $\dot{q}$, P_{total} , TCP $x/y/z$) and 161 tunable parameters.

3.3 XY10 Pick-and-Place Station

The XY10 station combines two stepper-driven linear axes (X, Y), a pneumatic Z axis and a vacuum gripper. Its model retains the basic dynamics required at the line-orchestration level (time-to-position, current draw per motion) but does not attempt to reproduce the chopper-mode switching pattern visible in the TDMS measurement, which would require a separate fast-electronics model of the drive. The FMU exposes axis positions, current draws and the vacuum gripper state.

4. Validation Against Field Measurements

Validation uses two TDMS recording sessions of the production line (first.tdms, 444 s and four.tdms, 287 s; 6 kHz sample rate). The voltage channels U0, U1 are clean 50 Hz sinusoids of peak ≈ 345 V (244 V RMS), used to calibrate the model supply. The current channels carry motor controller line-side currents, which contain the 50 Hz fundamental plus rectifier and PWM-switching harmonics. A narrow band-pass filter (40–60 Hz, 4th-order Butterworth, zero-phase) extracts the fundamental for direct comparison with the FMU's i_{phase_inst} output.

Key validation outcomes:

- **Voltage waveform:** exact agreement (sinusoid amplitude and frequency);
- **Conveyor current 50 Hz fundamental:** measured 1.99 A vs. simulated 2.30 A ($\approx 86\%$);
- **Phase:** cross-correlation alignment shift ≈ 3.7 ms ($\approx 18^\circ$ of the 20 ms cycle);
- **Steady-state RMS:** 1.06–1.17 A measured vs. 1.63 A simulated (FMU isolates conveyor alone; TDMS aggregates other line loads);
- **FMU-to-Dymola:** 0.00 % disagreement on all preserved outputs.

The 14 % residual on the fundamental amplitude is consistent with three modelling choices: (i) the constant mechanical load (5 N) is a simplification of the time-varying belt friction; (ii) the model treats the motor as directly grid-connected, whereas the real motor sits behind a diode-rectifier / PWM-inverter that introduces phase shifting and re-distribution of fundamental current; (iii) motor electrical parameters are nameplate values, not formally identified. Each of these is an explicit candidate for the parameter-identification campaign planned for the second half of the internship.

5. Energy Analysis

Running the orchestrated line model for a full operating cycle and integrating instantaneous power per subsystem yields a clear consumption hierarchy: the conveyor accounts for approximately 90 % of the line's electrical energy, primarily because of its continuous duty (running $\approx 70\%$ of cycle time at 108 W) compared with the intermittent stations. The XY10 station and CR5 cobot contribute the remaining 10 %, dominated by acceleration bursts. This finding has direct implications for predictive maintenance and Industry 5.0 KPIs: the conveyor motor is both the highest-energy consumer and the component most likely to benefit from thermal-state monitoring, now exposed in the v3 model as T_{motor} .

6. Specification Compliance and Remaining Work

Specification item	Status	Notes
Mechanical models (per subsystem)	Complete	Conveyor, CR5, XY10
Electrical models (per subsystem)	Complete	AC current waveform validated
Thermal models	Complete	Conveyor v3, lumped 1-state
Validation against TDMS	Complete	v7 figure, fundamental extraction
Energy / KPI analysis	Complete	90 % conveyor finding
Parameter identification / observability	Complete	Foundation laid; DoE planned
Spatial representation	In progress	Drafting 21–22 May
Coupling effects analysis	Open	Matrix to follow spatial diagram
Final report + presentation slides	To do	24 May – 11 June

Estimated overall completion of the technical scope: ~ 88 %. Including the final report and presentation slides as required deliverables brings the overall internship completion to approximately 70 %, with sufficient margin to absorb the spatial representation and coupling analysis before the 30 May semester review.

7. Next Steps

- **21–22 May:** Spatial representation diagram (2-D top-down layout of the ALIX line) and coupling matrix tabulating vibration, thermal and electrical interactions between subsystems.
- **23–24 May:** Documentation cleanup, figure organisation and report writing.
- **25–29 May:** Slide preparation and rehearsal for the 30 May semester review.
- **4 June:** On-site meeting at Dassault Systèmes, Paris.
- **11 June:** Final meeting with industrial supervisor.
- **16 June:** Internship end; final report submitted.

8. Conclusion

All three physical domains required by the specification — mechanical, electrical and thermal — are now implemented and validated for the conveyor subsystem, with the second station (CR5 cobot) complete in mechanical form and the third (XY10) in mechanical-plus-electrical form. End-to-end FMU validation, against both analytical predictions and TDMS field measurements, gives results well within the agreement tolerances expected for a first-pass digital twin. The remaining work before the semester review is the spatial-and-coupling deliverable, scheduled for the next two working days. The internship is on track for a complete defence on 11 June.